Sequence and Series - 1

Sequence : Arrangement of numbers in a logical pattern.

Eg 1, 2, 4, 8, 16 - - - (Term getting doubled)
1, 4, 7, 10, 13 - - - - (Term + 3)
1, 4, 9, 16, 25 - - - - (Sq. of Terms)

Series : Sum of all the elements of a sequence.

Eg; 1+2+4+8+16+ - - - -
1+4+7+10+13+ - - - -

Arithmetic Progression :

↳ Sequence whose terms increase or decrease by a fixed Number.

Eg: 1, 4, 7, 10, 13 - - - (+3 each) d=3 Increasing AP
4, 2, 0, -2, -4 - - - (-2 each) d=-2 Decreasing AP
3, 3, 3, 3, 3 - - - (±0 each) d=0 Constant AP

* N^{th} term of AP ⇒ $\boxed{T_n = a + (n-1)d}$

Shortcut-1 :
In an AP if $T_m = n$ and $T_n = m$, then $\boxed{T_{m+n} = 0}$

Eg If $T_5 = 6$ & $T_6 = 5$ then Find T_{11}?

Sol^n ∴ $T_{11} = T_{(6+5)} = 0$

Shortcut-2 :
If, $T_m = n$ and $T_n = m$ ⇒ $\boxed{T_r = m+n-r}$

Eg: $T_{64} = 13$ & $T_{13} = 64$ of an AP, find T_{100}

Sol^n $T_{100} = 64 + 13 - 100$
$= -23$ Ans

Shortcut-3 :
If $n \cdot T_n = m \cdot T_m$ ⇒ $\boxed{T_{m+n} = 0}$

Shortcut-4 :
If $T_9 = 37$ and $T_{15} = 65$, find 'd'.

Sol^n $d = \frac{65-37}{15-8} = \frac{28}{7} = 4$

i.e $\boxed{d = \frac{a_m - a_n}{m-n}}$ where $m > n$

Shortcut-5 :
How many 3-digit numbers divisible by 6?

Sol^n Think of between what numbers do our required no. would lie

100 - - - - - - 999 ⇒ 16. . . . 166.
(÷6) (÷6) ⇓ Take diff.
Divide them by 6 ⇒ n = 150 Ans

Ques How many no. b/w 11 to 90 are divisible by 7?

Sol^n $\frac{11}{7}$ - - - $\frac{90}{7}$ ⇒ 1. . . . 12. . ⇒ n = 11 Ans

Sum of n-terms of an AP :→

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$S_n = \frac{n}{2}[\text{First term} + \text{Last term}]$$

Shortcut-6 :
If value of sum of terms of an AP are independent of AP (i.e AP is not specified) then

Assume, AP : 1, 1, 1, 1, 1 - - -
⇓
AP : 1, 2, 3, 4, 5 - - -

Ques S_1 → Sum of 2n tems of AP, S_2 → Sum of 4n terms of AP } $S_2 - S_1 = 1000$
Find S_6 = 6n terms.

Sol^n Let, AP : 1, 1, 1, 1 - - - -

$S_1 = 2n$ and $S_2 = 4n$ ⇒ $S_2 - S_1 = 1000$
$4n - 2n = 1000$
⇒ n = 500

So, $S_6 = 6n = 6(500)$
$= 3000$ Ans

Ques If S_n denotes sum of first n terms of an AP then $\frac{S_{3n} - S_{n-1}}{S_{2n} - S_{2n-1}} = ?$

(A) 2n-1 (1) (B) 2n (2) (C) 2n+1 (3) (D) n/2 (1/2)

Let n = 1 and AP : 1, 1, 1, 1 - - -

So, $\frac{S_{3n} - S_{n-1}}{S_{2n} - S_{2n-1}} = \frac{S_3 - S_0}{S_2 - S_1} \Rightarrow \frac{3-0}{2-1}$
$= 3$ (C) Ans

Shortcut-7 :
If sum is given, then we would be asked ratio of nth terms or vice versa. Then use 'n → 2n-1' replace

$\frac{S_n}{S'_n} \rightarrow \frac{a_n}{a'_n}$
$n \rightarrow \frac{n+1}{2}$ replace

Ques For two APs, S_n are in ratio (3n+8) & (7n+15). Find ratio of 12th terms.

Sol^n $\frac{S_n}{S'_n} = \frac{3n+8}{7n+15}$; $\frac{a_{12}}{a'_{12}} = \frac{3[2n-1]+8}{7[2n-1]+15}$

$\frac{a_{12}}{a'_{12}} = \frac{3(23)+8}{7(23)+15}$ [Put n=12]

⇒ $\boxed{\frac{a_{12}}{a'_{12}} = \frac{7}{16}}$ Ans

Ques Ratio of nth terms of two APs is (2n+8) : (5n-3), find $\frac{S_{29}}{S'_{29}}$.

Sol^n $\frac{a_n}{a'_n} = \frac{2n+8}{5n-3} \longrightarrow \frac{S_{29}}{S'_{29}} = \frac{2\left(\frac{n+1}{2}\right)+8}{5\left(\frac{n+1}{2}\right)-3}$

⇒ $\frac{S_{29}}{S'_{29}} = \frac{2(15)+8}{5(15)-3} \Rightarrow \frac{38}{72} \Rightarrow \frac{19}{36}$ Ans

Shortcut-8 : Common terms of 2 APs

* New AP of Common terms of AP_1 & AP_2 would have d as
$\boxed{d = LCM(d_1, d_2)}$ (AP)
* Common term will remain greatest term, AP_1 aur AP_2 mein se smaller last term ke less than equal to hogi.

Ques Find No. of terms common b/w 3, 7, 11 - - - 407 & 2, 9, 16 - - - 709.

Sol^n 3, 7, 11, 15, 19, (23) - - - 407
2, 9, 16, (23) 30 - - - - - 709
(First common term)

d = LCM (4, 7)
d = 28 ; a = 23

$a_n \leq 407$ (Smaller Term)
$a + (n-1)d \leq 407$
$23 + (n-1)28 \leq 407$
⇒ $n \leq 14.7$
i.e n = 14 Ans

Highlights of an AP :→

1) If an AP → +k, −k, ×k, ÷k (where k is some constant) then resulting sequence is also an AP.

2) To assume terms in an AP
3 terms ⇒ (a-d) (a) (a+d)
4 terms ⇒ (a-3d) (a-d) (a+d) (a+3d)
5 terms ⇒ (a-2d) (a-d) (a) (a+2d) (a+2d)
6 terms ⇒ (a-5d) (a-3d) (a-d) (a+d) (a+3d) (a+5d)

3) Sum of two terms equidistant from beginning and end is constant and equal to sum of first and last term.

i.e $a_3 + a_{11} = a_4 + a_{10} = a_5 + a_9$
→ Sum of subscripts should be same
→ Only for sum of two terms.

4) For any series (AP, GP, etc)
$\boxed{T_n = S_n - S_{n-1}}$

5) If a, b & c are in AP ⇒ $\boxed{2b = a+c}$

Ques If $\frac{1}{b+c}$, $\frac{1}{c+a}$, $\frac{1}{a+b}$ are in AP, then prove that

$\frac{a}{b+c}, \frac{b}{a+c}, \frac{c}{a+b}$ are also in AP.

Sol^n Any algebraic operation of AP by some const gives us AP only. So

$AP \rightarrow \frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$

$\times(a+b+c)$ ↳ $\frac{a+(b+c)}{b+c}, \frac{b+(c+a)}{c+a}, \frac{c+(a+b)}{a+b}$

-1 ↳ $\frac{a}{b+c}+\frac{1}{-1}, \frac{b}{c+a}+\frac{1}{-1}, \frac{c}{a+b}+\frac{1}{-1}$

$\Rightarrow \boxed{\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}} \rightarrow$ AP, Hence Proved

Ques $a_1, a_2, a_3 \cdots a_n$ are in AP and, if

$\frac{1}{a_1 a_n} + \frac{1}{a_2 a_{n-1}} + \frac{1}{a_3 a_{n-2}} \cdots \frac{1}{a_n a_1} = \frac{\lambda}{a_1 a_n}\left[\frac{1}{a_1} + \cdots \frac{1}{a_n}\right]$

then find λ.

Sol^n Multiply divide by (a_1+a_n) and since terms are in AP

$a_1+a_n = a_2+a_{n-1} = a_3+a_{n-2} = \cdots$

$\Rightarrow \frac{1}{a_1+a_n}\left[\frac{a_1+a_n}{a_1 a_n} + \frac{a_2+a_{n-1}}{a_2 \cdot a_{n-1}} + \cdots \frac{a_1+a_n}{a_n \cdot a_1}\right]$

$= \frac{1}{a_1+a_n}\left[\frac{1}{a_1}+\frac{1}{a_n} + \frac{1}{a_2}+\frac{1}{a_{n-1}} + \cdots \frac{1}{a_n}+\frac{1}{a_1}\right]$ (Terms Repeating Twice)

$= \frac{2}{a_1+a_n}\left[\frac{1}{a_1}+\frac{1}{a_2}+\frac{1}{a_3} \cdots \frac{1}{a_{n-1}}+\frac{1}{a_n}\right]$

$\Rightarrow \boxed{\lambda = 2}$ Ans

Ques If sum of first 40 terms of $3+4+8+9+13+14+18+19 \cdots$ is $(102)m$, find m.

[Clubbing two adjacent terms, gives us an AP]

Sol^n $S_n = 3+4+8+9+13+14+18+19 \cdots$

$S_n = 7+17+27+37 \cdots$

↳ since two terms clubbed together, we now find S_{20} instead of S_{40}.

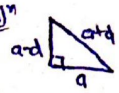
$\Rightarrow S_{20} = \frac{20}{2}[2(7)+(20-1)10]$

$102m = 10[14+190] \Rightarrow 2040$

$\boxed{m = 20}$ Ans

Ques Sides of right angled Δ are in AP. If area is 24, find length of smallest side.

Sol^n



Area $= \frac{1}{2}(a)(a-d) = 24$

$\boxed{a(a-d) = 48}$ —(1)

By Pythagorous Theorem: $(a+d)^2 = (a-d)^2 + a^2$

$4d(3d) = 48$

$12d^2 = 48 \Rightarrow \boxed{d=2}$

$\Rightarrow 4ad = a^2 \Rightarrow \boxed{a = 4d}$

Put eq(1) ↙

So, $a = 8$ & Smallest side $(x) = a-d$

$\Rightarrow \boxed{x = 6}$ Ans

Geometric Progression :

↳ Sequence in which ratio of two consecutive terms is constant.

Eg: $1 \xrightarrow{\times 3} 3 \xrightarrow{\times 3} 9 \xrightarrow{\times 3} 27 \xrightarrow{\times 3} 81, \cdots$

nth term of GP :→

$a, ar, ar^2, ar^3, ar^4 \cdots$

$\boxed{T_n = ar^{n-1}}$ $r>1 \Rightarrow$ Rising GP; $r=1 \Rightarrow$ Constant GP; $r<1 \Rightarrow$ Falling GP

NOTE: $2, 2, 2, 2, 2 \cdots$

$(d=0)$ & $(r=1) \Rightarrow$ Hence, it is constant GP as well as constant AP.

Ques In an Rising GP, sum of 2nd & 6th term is $\frac{25}{2}$, and product of 3rd & 5th term is 25. Find sum of 4th, 6th & 8th terms.

Sol^n $ar + ar^5 = \frac{25}{2}$; $(ar^2)(ar^4) = 25$

$a(r+r^5) = \frac{25}{2}$; $\boxed{ar^3 = 5}$

$\frac{5}{r^3}(r+r^5) = \frac{25}{2} \cdot 5 \Rightarrow \frac{1+r^4}{r^2} = \frac{5}{2}$

$2+2r^4-5r^2 = 0 \Rightarrow (2r^2-1)(r^2-2) = 0$

So, $r^2 = \frac{1}{2}$ (x) or 2 (✓) because Rising GP

Now,

$T_4+T_6+T_8 = ar^3 + ar^5 + ar^7$

$= 5 + (ar^3)[r^2+r^4]$

$= 5 + 5[2+4] \Rightarrow 35$ Ans

Sum of n-terms of GP :→

$\boxed{S_n = \frac{a(r^n-1)}{r-1} = \frac{a(1-r^n)}{1-r}}$; $\boxed{S_\infty = \frac{a}{1-r}}$ Only when $|r|<1$ & $-1<r<1$

Ques If $0<\theta, \phi<\pi/2$, $x = \sum_{n=0}^{\infty}\cos^{2n}\theta$, $y = \sum_{n=0}^{\infty}\sin^{2n}\phi$ & $z = \sum_{n=0}^{\infty}\cos^{2n}\theta \cdot \sin^{2n}\phi$ then find relation b/w x, y & z.

Sol^n $x = 1+\cos^2\theta+\cos^4\theta+\cdots$

$x = S_\infty = \frac{1}{1-\cos^2\theta} \Rightarrow \boxed{x = \frac{1}{\sin^2\theta}}$

$y = 1+\sin^2\phi+\sin^4\phi+\cdots$

$y = S_\infty = \frac{1}{1-\sin^2\phi} \Rightarrow \boxed{y = \frac{1}{\cos^2\phi}}$

And,

$z = 1+\cos^2\theta\cdot\sin^2\phi+\cos^4\theta\cdot\sin^4\phi+\cdots$

$z = \frac{1}{1-\cos^2\theta\cdot\sin^2\phi} \Rightarrow \frac{1}{1-(1-\sin^2\theta)(1-\cos^2\phi)}$

$z = \frac{1}{1-\left(1-\frac{1}{x}\right)\left(1-\frac{1}{y}\right)} \Rightarrow \frac{1}{z} = 1-\left(1-\frac{1}{x}-\frac{1}{y}+\frac{1}{xy}\right)$

$\Rightarrow \frac{1}{z} = \frac{1}{x}+\frac{1}{y}-\frac{1}{xy} \Rightarrow \frac{xy+z}{xyz} = \frac{x+y}{xy}$

$\Rightarrow \boxed{z(x+y) = xy+z}$ Ans

Ques The product of $2^{1/4} \cdot 4^{1/16} \cdot 8^{1/48} \cdots \infty$ is

Sol^n $S = 2^{1/4} \cdot 2^{1/8} \cdot 2^{1/16} \cdots$

$S = 2^{\left(\frac{1/4}{1-1/2}\right)} \Rightarrow 2^{\left(\frac{1}{4}\times\frac{2}{1}\right)}$

$\Rightarrow \boxed{S = \sqrt{2}}$ Ans

Highlights of GP :→

1) If each term of a GP is multiplied or divided by same number, then resulting sequence is also a GP.

$a, ar, ar^2, ar^3 \cdots$ ↳ $\times K \Rightarrow$ GP; ↳ $\div K \Rightarrow$ GP

2) If each term of GP raised to same power, then resulting sequence is also a GP.

$1, 2, 4, 16, 32 \cdots \Rightarrow$ GP $(r=2)$

$1^3, 2^3, 4^3, 16^3, 32^3 \cdots \Rightarrow$ GP $(r=2^3)$

3) To assume terms in GP

3 terms $\Rightarrow \frac{a}{r}, a, ar$

4 terms $\Rightarrow \frac{a}{r^3}, \frac{a}{r}, ar, ar^3$

5 terms $\Rightarrow \frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$

4) In a finite GP, product of terms equidistant from end & beginning are equal.

$(a_1a_n = a_2 \cdot a_{n-1} = a_3 \cdot a_{n-2} = \cdots)$

5) If a, b, c are in GP $\Rightarrow \boxed{b^2 = ac}$

Ques The sum of first 4 terms of GP is $65/12$ and the sum of their respective reciprocals is $65/18$. If product of first three terms is 1, and third term is a, find $2a$.

Sol^n $a(1+r^2+r+r^3) = 65/12$

& $\frac{1}{a}+\frac{1}{ar}+\frac{1}{ar^2}+\frac{1}{ar^3} = 65/18$

$\frac{(r^3+r^2+r+1)}{ar^3} = \frac{65}{18}$

$\Rightarrow \frac{65}{12a^2r^3} = \frac{65}{18} \Rightarrow \boxed{a^2r^3 = \frac{3}{2}}$

Also, it is given

$a \cdot ar \cdot ar^2 = 1 \Rightarrow (ar)^3 = 1$

$\Rightarrow \boxed{a = 1/r}$

So, $a^2r^3 \Rightarrow \boxed{r = 3/2}$, $\boxed{a = 2/3}$ $\Rightarrow 2a = 2ar^2 = 2\left(\frac{2}{3}\right)\left(\frac{3}{2}\right)^2$

$\Rightarrow \boxed{2a = 3}$ Ans

Sequence Convertible to GP :

Type-01 : Converting Recurring Decimal to Fraction :→

$x = 0.4\overline{23}$

Sequence & Series - 2

Sol^n $x = 0.\overline{42323}23\ldots$

$x = 0.4 + 0.023 + 0.00023 + \ldots$

$x = \frac{4}{10} + \left(\frac{23}{10^3} + \frac{23}{10^5} + \frac{23}{10^7} + \ldots\right)$

$x = \frac{4}{10} + \left(\frac{23/10^3}{1 - 1/100}\right) \Rightarrow \frac{4}{10} + \frac{23}{990}$

~~$x = \frac{396+23}{990} \Rightarrow \frac{036}{990} \frac{429}{990}$ Ans~~

$x = \frac{396+23}{990} \Rightarrow \frac{419}{990}$ Ans

* Type-02 :→

Ques If $9 + 99 + 999 + \ldots$ upto 49 terms $= 10\frac{(10^{\lambda}-1)}{M} - 49$, then find $(\lambda+M)$.

Sol^n $S = 9 + 99 + 999 + \ldots$

$S = (10-1) + (100-1) + (1000-1) + \ldots$

$S = (10 + 10^2 + 10^3 \ldots) - 49$

$S = \frac{10(10^{49}-1)}{9} - 49 \Big]_{M=9}^{\lambda=49} \Rightarrow 58$ Ans

Ques Find sum of first 20 terms of sequence $0.7, 0.77, 0.777, \ldots$

Sol^n $S = 7\left(\frac{1}{10} + \frac{11}{100} + \frac{111}{1000} + \ldots\right)$

~~$S = 7\left(\frac{10-1}{10} + \frac{10+1}{\cdot}\right.$~~

$S = \frac{7}{9}\left(\frac{9}{10} + \frac{99}{100} + \frac{999}{1000} + \ldots\right)$

$S = \frac{7}{9}\left[\frac{10-1}{10} + \frac{100-1}{100} + \frac{1000-1}{1000} + \ldots\right]$

$S = \frac{7}{9}\left[1 - \frac{1}{10} + 1 - \frac{1}{100} + 1 - \frac{1}{1000} + \ldots\right]$

$S = \frac{7}{9}\left[(20) - \left(\frac{\frac{1}{10}\left[1-\left(\frac{1}{10}\right)^{20}\right]}{1-\frac{1}{10}}\right)\right]$

$S = \frac{7}{9}\left[20 - \frac{1}{9}\left(1 - \left(\frac{1}{10}\right)^{20}\right)\right]$

$S = \frac{7}{9}\left[20 - \frac{1}{9} + \frac{1}{9}\left(\frac{1}{10}\right)^{20}\right] \Rightarrow \frac{7}{9}\left[\frac{179}{9} - \frac{1}{9}\left(\frac{1}{10}\right)^{20}\right]$ Ans

* Type-03 :→

Ques Find the sum

$S = (x+y) + (x^2+xy+y^2) + (x^3+x^2y+xy^2+y^3) + \ldots$ n terms

Sol^n $\frac{x^2-y^2}{x-y} = x+y$; $\frac{x^3-y^3}{x-y} = (x^2+xy+y^2)$

$S = \frac{x^2-y^2}{x-y} + \frac{x^3-y^3}{x-y} + \frac{x^4-y^4}{x-y} \ldots$ n terms

$S = \frac{1}{x-y}\left[(x^1+x^2\ldots \text{n terms}) - (y^1+y^2\ldots \text{n terms})\right]$

$S = \frac{1}{x-y}\left[\frac{x^1(x^n-1)}{x-1} - \frac{y^1(y^n-1)}{y-1}\right]$ Ans

* Type-04 :→

Ques $S = \frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \ldots$ n terms

Sol^n $S = \left(1-\frac{1}{2}\right) + \left(1-\frac{1}{4}\right) + \left(1-\frac{7}{8}\right) \ldots$ n terms

$S = n - \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} \ldots \text{n terms}\right)$

$S = n - \frac{\frac{1}{2}\left(\frac{1}{2^n}-1\right)}{1-\frac{1}{2}} \Rightarrow n - \left(1 - \frac{1}{2^n}\right)$ Ans

Arithmetic-Geometric Progression (AGP) :

→ Sequences in which one part of terms would be in AP while other part would be in GP.

Sum of n & Infinite terms of AGP :→

↳ Sequence of common ratio se multiply Karo aur ek place shift Karke ~~add~~ minus Kara

Eg: $S = \underset{\times 3}{\frac{3}{5}} + \underset{\times 3}{\frac{5}{15}} + \underset{\times 3}{\frac{7}{45}} + \frac{9}{135} + \ldots$ (+2, +2, +2)

So, $S = \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \ldots$

$\frac{1}{3}S = \frac{3}{15} + \frac{5}{45} + \frac{7}{135} + \ldots$

$\frac{2}{3}S = \frac{3}{5} + \left(\frac{2}{15} + \frac{2}{45} + \frac{2}{135} + \ldots\right)$

$S = \frac{9}{10} + \frac{3}{2}\left[\frac{\left(\frac{2}{15}\right)}{1-\frac{1}{3}}\right] \Rightarrow \frac{9}{10} + \left[\frac{1}{5}\right]\frac{3}{2}$

$S = \frac{14}{10} \frac{12}{10} \Rightarrow 1.2$ Ans

Ques Find $S = 1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \frac{14}{3^4} \ldots$

Sol^n $S = 1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \frac{14}{3^4} \ldots$

$\frac{1}{3}S = \frac{1}{3} + \frac{2}{3^2} + \frac{6}{3^3} + \frac{10}{3^4} \ldots$

$\frac{2}{3}S = 1 + \frac{1}{3} + \left(\frac{4}{3^2} + \frac{4}{3^3} + \frac{4}{3^4} \ldots\right)$

$\frac{2}{3}S = \frac{4}{3} + \frac{4/3^2}{1-1/3} \Rightarrow \frac{4}{3} + \frac{2}{3}$

$S = \frac{6}{2} \times \frac{3}{2} \Rightarrow 3$ Ans

Miscellaneous Series :

* Type-01 : Sequences adding up to standard Summations :→

- $\Sigma n = \frac{n(n+1)}{2}$
- $\Sigma 1 = n$
- $\Sigma n^2 = \frac{n(n+1)(2n+1)}{6}$
- Σ(even no.) $= n(n+1)$
- $\Sigma n^3 = \left(\frac{n(n+1)}{2}\right)^2$
- Σ(odd no.) $= n^2$

Ques If S_{10} of $\left(1\frac{3}{5}\right)^2 + \left(2\frac{2}{5}\right)^2 + \left(3\frac{1}{5}\right)^2 + 4^2 + \left(4\frac{4}{5}\right)^2 + \ldots$ gs $\frac{16}{5}m$, find m.

Sol^n $S_n = \left(\frac{8}{5}\right)^2 + \left(\frac{12}{5}\right)^2 + \left(\frac{16}{5}\right)^2 + \left(\frac{20}{5}\right)^2 + \ldots$

$S_{10} = \frac{4^2}{5^2}\left[\underset{+1^2}{2^2} + 3^2 + 4^2 \ldots \underset{-1^2}{+11^2}\right]$

$S_{10} = \frac{16}{25} \times \left[\frac{10(11)(21)}{2} \frac{11\left(\frac{2}{12}\right)(23)}{6} - 1\right]$

$S_{10} = \frac{16}{25}\left[506 - 1\right] \Rightarrow \frac{16}{25_5} \times 505^{101}$

$\Rightarrow \boxed{m = 101}$ Ans

Ques Find S_9 of $\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + \ldots$

Sol^n $T_r = \left(\frac{1^3+2^3+3^3 \ldots r \text{ terms}}{1+3+5+7 \ldots r \text{ terms}}\right)$

$\sum_{r=1}^{9} T_r = \frac{\left[\frac{r(r+1)}{2}\right]^2}{r^2} \Rightarrow \frac{\left[\frac{9(9+1)}{2}\right]^2}{9^2}$

$S_9 = \sum_{r=1}^{9} \frac{(r+1)^2}{4} \Rightarrow \frac{1}{4}\left[\underset{+1^2}{2^2} + 3^2 + 4^2 \ldots \underset{-1^2}{10^2}\right]$

$\Rightarrow S_9 = \frac{1}{4} \times \left[\frac{10(11)(21)}{6} - 1\right] \Rightarrow \frac{1}{4} \times \frac{2304}{6}$

$\boxed{S_9 = \frac{576}{6}}$ $\Rightarrow \boxed{S_9 = 96}$ Ans

Ques If α, β are natural number such that $100\alpha - 199\beta = (100)(100) + (99)(101) + (98)(102) + \ldots (1)(199)$, then find slope of line passing through (α, β) & origin.

Sol^n $S = (100)(100) + (99)(101) + \ldots (1)(199)$

$T_r = (100-r)(100+r) \Rightarrow 100^2 - r^2$

$S = \sum_{r=0}^{99}\left[100^2 - r^2\right] \Rightarrow \sum_{r=0}^{99}(100)^2 - \sum_{r=0}^{99} r^2$

$S = 100(100)^2 - \frac{99(100)(199)}{6}$

$S = (100)^3 - \left(\frac{99 \times 100}{6}\right)199 = 100\alpha - 199\beta$

So, $\alpha = 3$ & $\beta = 1650$

And, slope $(m) = \frac{\beta - 0}{\alpha - 0} = \frac{1650}{3}$

$\Rightarrow \boxed{m = 550}$ Ans

* Type-02 : Continued Product :→

Eg: $S = 1.2 + 2.3 + 3.4 + \ldots + n(n+1)$

Method-1 : Sigma Method ⇒ Write general term and apply sigma.

$S = \sum_{r=1}^{n} r(r+1) \Rightarrow \sum_{r=1}^{n} r^2 + \sum_{r=1}^{n} r$

$S = \frac{n(n+1)(n+2)}{6} + \frac{n(n+1)}{2} \Rightarrow \frac{n(n+1)}{2}\left[\frac{n+2}{3} + 1\right]$

$\boxed{S = \frac{n(n+1)(n+2)}{3}}$ Ans

Method-2 : Method of Difference ⇒ Ek baad wali aur ek pehle wali term ke diff. se GT Ko multiply Karo

Eg $T_r = \frac{1}{3} r(r+1)\left[(r+2) - (r-1)\right]$

$T_r = \frac{1}{3}\left[r(r+1)(r+2) - (r-1)r(r+1)\right]$

$T_1 = \frac{1}{3}\left[1\cdot2\cdot3 - 0\cdot1\cdot2\right]$

$T_2 = \frac{1}{3}\left[2\cdot3\cdot4 - 1\cdot2\cdot3\right]$

$T_3 = \frac{1}{3}\left[3\cdot4\cdot5 - 2\cdot3\cdot4\right]$

$\vdots$

$T_n = \frac{1}{3}\left[n(n+1)(n+2) - (n-1)(n)(n+1)\right]$ (+)

$S_n = \frac{1}{3}\left[n(n+1)(n+2) - 0\right]$

$\Rightarrow \boxed{S_n = \frac{n(n+1)(n+2)}{3}}$ Ans

Shortcut-9: Use only when continued product alone in numerator.

* Write General term and multiply it by 'EK baad wala' term.
* Divide General term by no. of terms present in Numerator after first step.

Ques $S = 1\cdot2\cdot3 + 2\cdot3\cdot4 + 3\cdot4\cdot5 + \ldots$ n terms

Sol^n $T_n = n(n+1)(n+2)$

$\boxed{S_n = \frac{n(n+1)(n+2)(n+3)}{4}}$ Ans

Ques $S = 1\cdot2\cdot3\cdot4 + 2\cdot3\cdot4\cdot5 + \ldots$ n terms

$\boxed{S = \frac{n(n+1)(n+2)(n+3)(n+4)}{5}}$ Ans

* Type-03: Using method of difference :→

We apply 'Method of difference' for series whose 1st or 2nd diff. are in AP.

Eg: $6 + 13 + 22 + 33 + \ldots$ n terms

$7 \quad 9 \quad 11$ ← AP of $d=2$

So,

$S = 6 + 13 + 22 + 33 + \ldots + T_n$

$(-)\ S = \quad 6 + 13 + 22 + \ldots + T_{n-1} + T_n$

$0 = 6 + (7 + 9 + 11 + \ldots \text{ } (n-1)^{th} \text{ term}) - T_n$

$\Rightarrow T_n = 6 + \frac{n-1}{2}\left[2(7) + (n-2)2\right]$

$\Rightarrow T_n = 6 + (n-1)(n-5)$

$\Rightarrow \boxed{T_n = n^2 + 1n + 1}$ → Apply Σ & solve by standard summations

Shortcut-10: We can directly write General term T_n for such series.

(i) Take differences until you get 0

(ii) From 1st diff., start multiplying the differences by $(n-1), (n-1)(n-2), \ldots$ and divide them by (no. of new terms multiplied)!

(iii) Add all the differences after doing this

Eg $6 + 13 + 22 + 33 + \ldots$

$7 \quad 9 \quad 11$

$2 \quad 2$

0

$T_n = 6 + \frac{7(n-1)}{1!} + \frac{2(n-1)(n-2)}{2!} + \frac{0(n-1)(n-2)(n-3)}{3!}$

$\Rightarrow T_n = 6 + 7(n-1) + (n-1)(n-2)$

$\Rightarrow \boxed{T_n = n^2 + 4n + 1}$

$\hookrightarrow S_n = \sum_{n=0}^{n} n^2 + 4\sum_{n=0}^{n} n + \sum_{n=0}^{n} 1$

Ans $\boxed{S_n = \frac{n(n+1)(2n+1)}{6} + \frac{4n(n+1)}{2} + n}$

Eg: $S = 2 + 5 + 10 + 17 + 26 + 37 + \ldots T_n$

$3 \quad 5 \quad 7 \quad 9 \quad 11$

$2 \quad 2 \quad 2 \quad 2$

$0 \quad 0 \quad 0$

$T_n = 2 + \frac{3(n-1)}{1!} + \frac{2(n-1)(n-2)}{2!}$

$\Rightarrow T_n = 2 + 3n - 3 + n^2 - 3n + 2$

$\boxed{T_n = n^2 + 1}$

$\hookrightarrow \boxed{S_n = \frac{n(n+1)(2n+1)}{6} + n}$ Ans

* Type-04: Splitting nth term as difference of roots (Continued product in Denominator) :→

Eg: $S = \frac{1}{1\cdot2} + \frac{1}{2\cdot3} + \frac{1}{3\cdot4} + \ldots + \frac{1}{n(n+1)}$

Sol^n Sabse chotf term aur sabse badf term ka difference Numerator mein likhdo.

$T_r = \frac{1}{r(r+1)} \Rightarrow \frac{(r+1)-r}{r(r+1)}$

$T_r = \frac{1}{r} - \frac{1}{r+1}$

$T_1 = 1 - \frac{1}{2}$

$T_2 = \frac{1}{2} - \frac{1}{3}$

$T_3 = \frac{1}{3} - \frac{1}{4}$

$\vdots$

(+) $T_n = \frac{1}{n} - \frac{1}{n+1}$

$S_n = 1 - \frac{1}{n+1} \Rightarrow \boxed{S_n = \frac{n}{n+1}}$ Ans

Shortcut-11: Use this when Continued product in Denominator.

(i) Eliminate smallest factor of Last term and Largest factor of first term.

(ii) Then, substract the remaining two terms after elimination.

(iii) Select any term of series, take diff. of smallest & largest factor and divide the above differences by 4b.

Eg: $S = \frac{1}{1\cdot2\cdot3} + \frac{1}{2\cdot3\cdot4} + \frac{1}{3\cdot4\cdot5} + \ldots$ n terms

Sol^n $S = \frac{1}{1\cdot2\cdot3} + \frac{1}{2\cdot3\cdot4} + \ldots \frac{1}{n(n+1)(n+2)}$

Diff = 2

$S = \frac{1}{2}\left[\frac{1}{1\cdot2} - \frac{1}{(n+1)(n+2)}\right]$ Ans

Eg: $S = \frac{1}{1\cdot2\cdot3\cdot4} + \frac{1}{2\cdot3\cdot4\cdot5} + \ldots + \frac{1}{n(n+1)(n+2)(n+3)}$

Diff = 3

Sol^n

$S = \frac{1}{3}\left[\frac{1}{1\cdot2\cdot3} - \frac{1}{(n+1)(n+2)(n+3)}\right]$

$\Rightarrow \boxed{S = \frac{1}{3}\left[\frac{1}{6} - \frac{1}{(n+1)(n+2)(n+3)}\right]}$ Ans

Harmonic Progression :

↳ Reciprocal of HP's terms are in AP.

i.e. $a, b, c, d \Rightarrow$ HP

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \frac{1}{d} \Rightarrow$ AP

Eg: If the 3rd, 6th and last term of a HP are $\frac{1}{3}, \frac{1}{5}$ & $\frac{3}{203}$, then find total no. of terms.

Sol^n Convert into AP and solve.

HP	AP
$a_3 \to 1/3$	$a_3 = 3$
$a_6 \to 1/5$	$a_6 = 5$
$a_n \to 3/203$	$a_n = \frac{203}{3}$

$\Rightarrow d = \frac{5-3}{6-3} \Rightarrow \frac{2}{3}$

$\Rightarrow a + 2d = 3$

$\boxed{a = 5/3}$

So, $a_n = a + (n-1)d$

$\frac{203}{3} = \frac{5}{3} + (n-1)\frac{2}{3} \Rightarrow 198 = (n-1)2$

$\Rightarrow \boxed{n = 100}$ Ans

Shortcut-12: In a AP, if $T_m = n$ & $T_n = m$ then,

$\boxed{T_{m+n} = \frac{mn}{m+n}}$

Ques Find the middle root, if roots of the eqn $x^3 - 11x^2 + 36x - 36 = 0$ are in HP.

Sol^n $x^3 - 11x^2 + 36x - 36 = 0$ $\begin{matrix}\alpha\\ \beta\\ \gamma\end{matrix}\Big\}$ in HP

To convert eqn into the one, whose roots are $\left(\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}\right)$, so that they come in AP, we need to replace $x \to \frac{1}{x}$

$\Rightarrow -\left(\frac{1}{x^3} - \frac{11}{x^2} + \frac{36}{x} - 36 = 0\right)$

$\Rightarrow 36x^3 - 36x^2 + 11x - 1 = 0$ — 1/α, 1/β, 1/γ

Let, $\left[\frac{1}{\alpha} = A - d,\ \frac{1}{\beta} = A,\ \& \ \frac{1}{\gamma} = A + d\right]$

So, Sum of roots $= -b/a$

$3A = 1 \Rightarrow \boxed{A = 1/3}$

$\hookrightarrow \frac{1}{\beta} = \frac{1}{3} \Rightarrow \boxed{\beta = 3}$ Ans

Ques Let +ve numbers a, b, c, d be in AP. Then abc, abd, acd, bcd are in HP.

Sol^n $\frac{1}{xabcd}$

$a, b, c, d \Rightarrow$ AP

$\hookrightarrow \frac{1}{bcd}, \frac{1}{acd}, \frac{1}{abd}, \frac{1}{abc} \Rightarrow$ AP

$\hookrightarrow \frac{1}{abc}, \frac{1}{abd}, \frac{1}{acd}, \frac{1}{bcd} \Rightarrow$ AP

So, abc, abd, acd, bcd } in HP

Arithmetic Mean (AM):

AM of 'a' & 'b' = $\frac{a+b}{2}$

AM of 'a' 'b' & 'c' = $\frac{a+b+c}{3}$

AM of n terms = $\frac{a_1+a_2+a_3+\cdots a_n}{n}$

To Insert 'n' AMs b/w a and b :→

$a \quad \underset{+d}{A_1} \quad \underset{+d}{A_2} \quad \underset{+d}{A_3} \quad \cdots\cdots \quad A_n \quad \underset{+d}{b}$

$$d = \frac{b-a}{n+1}$$

Sum of 'n' AMs b/w a and b :→

$$\sum_{i=1}^{n} A_i = n \times AM(a,b)$$

↳ First aur last term ka AM leke usko se multiply kardo.

Ques If eleven AMs are inserted b/w 28 and 10, then find the no. of integral AMs.

Sol'n $d = \frac{-28+10}{11+1} = \frac{-18}{12} \Rightarrow -1.5$

28 — — — — — — — — — — 10

−3 −3 −3 −3 −3

Integral AMs = 5 **Ans**

Ques If n arithmetic means are inserted b/w 2 and 38, then the sum of the resulting series is obtained as 200. Then find the value of n.

Sol'n $2 \quad A_1 \quad A_2 \quad A_3 \quad A_4 \quad \cdots \quad A_n \quad 38$

$A_1+A_2+\cdots A_n = \frac{n}{2}(2+38) \Rightarrow n \times 20 \Rightarrow 20n$

Also, $2+A_1+A_2+\cdots A_n+38 = 200$

$40+20n = 200 \Rightarrow \boxed{n=8}$ **Ans**

Ques Let a,b,c be +ve integers such that b/a is an integer. If a,b,c are in GP and AM of a,b,c is (b+2) then find $\frac{a^2+a-14}{a+1}$.

Sol'n Since a,b,c are in GP

$\Rightarrow \frac{b}{a} = r \Rightarrow$ +ve integer (as 'a' & 'b' both are +ve)

Also, AM = b+2

$\frac{a+b+c}{3} = b+2 \Rightarrow \frac{a+ar+ar^2}{3} = ar+2$

$a+ar+ar^2 = 3ar+6$

$ar^2-2ar+a = 6 \Rightarrow a(r^2-r+1) = 6$

$\Rightarrow \boxed{a = \frac{6}{r^2-r+1}} \Rightarrow \boxed{a = \frac{6}{(r-1)^2}}$

Since 'a' is a +ve integer

$r = 1 \longrightarrow a = 6/0$ ✗

$\boxed{r=2} \longrightarrow a = 6/1$ (✓)

So, $\frac{a^2+a-14}{a+1} = \frac{36+6-14}{6+1} \Rightarrow \frac{28}{7}$

$\boxed{\frac{a^2+a-14}{a+1} = 4}$ **Ans**

Geometric Mean : (GM)

GM of 'a' & 'b' = $(ab)^{1/2}$

GM of 'a', 'b' & 'c' = $(abc)^{1/3}$

GM of n terms = $(a_1 \cdot a_2 \cdot a_3 \cdots a_n)^{1/n}$

To Insert 'n' GMs b/w a and b :→

$a \quad \underset{\times r}{G_1} \quad \underset{\times r}{G_2} \quad \cdots \quad G_n \quad \underset{\times r}{b}$

$$r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$$

Product of 'n' GMs b/w a and b :→

$$\prod_{i=1}^{n} G_i = [GM(a,b)]^n$$

↳ First aur last term ka GM leke usko itne GM hai (n), utne baar multiply kardo.

Harmonic Mean (HM):

HM of 'a' & 'b' = $\frac{2}{1/a+1/b}$

HM of 'a' 'b' & 'c' = $\frac{3}{1/a+1/b+1/c}$

HM of n terms = $\frac{n}{1/a_1+1/a_2+\cdots 1/a_n}$

To Insert n-HMs b/w a and b :→

↳ Can't be inserted directly.

$a, \quad H_1, \quad H_2, \quad \cdots\cdots \quad H_n, \quad b$

$\frac{1}{a}, \quad \underset{+d}{\frac{1}{H_1}}, \quad \underset{+d}{\frac{1}{H_2}}, \quad \cdots\cdots \quad \underset{+d}{\frac{1}{H_n}}, \quad b$

$$d = \frac{\frac{1}{b}-\frac{1}{a}}{n+1}$$

↳ Derived from common diff. of AMs

Sum of Reciprocals of n HM :→

$$\sum_{i=1}^{n} \frac{1}{H_i} = \frac{n}{HM(a,b)}$$

Ques If 9 AM and HM are inserted b/w 2 & 3, prove that $A + \frac{6}{H} = 5$.

Sol'n $2, A_1, A_2, \cdots A_9, 3$

$2, H_1, H_2, \cdots H_9, 3$

↳ $\frac{1}{2}, \frac{1}{H_1}, \frac{1}{H_2}, \cdots \frac{1}{H_9}, \frac{1}{3}$

$d_1 = \frac{\frac{1}{b}-\frac{1}{a}}{n+1} = \frac{\frac{3-2}{a+1}}{9+1} = \frac{1}{10}$

$d_2 = \frac{\frac{1}{b}-\frac{1}{a}}{n+1} = \frac{\frac{1}{3}-\frac{1}{2}}{9+1} \Rightarrow \frac{-1}{60}$

Now,

$A_1 = 2+\frac{1}{10} = \frac{21}{10}$

$\frac{1}{H_1} = \frac{1}{2}+\left(\frac{-1}{60}\right) = \frac{29}{60}$

So, $A_1 + 6\left(\frac{1}{H_1}\right) = \frac{21}{10} + 6\left(\frac{29}{60}\right)$

$= 5$

Hence Proved

Ques If A_1, A_2 ; G_1, G_2 ; & H_1, H_2 are means b/w two quantities, prove that

$$\frac{G_1G_2}{H_1H_2} = \frac{A_1+A_2}{H_1+H_2}.$$

Sol'n To prove : $\frac{H_1+H_2}{H_1H_2} = \frac{A_1+A_2}{G_1G_2}$

LHS : $\frac{1}{H_2}+\frac{1}{H_1} = \frac{n(a+b)}{2ab}$

RHS : $\frac{A_1+A_2}{G_1G_2} = \frac{n\left(\frac{a+b}{2}\right)}{ab}$

⇒ Hence Proved

Relation b/w AM, HM & GM :

1) $\boxed{G^2 = AH}$

2) $\boxed{AM \geq GM \geq HM}$

↳ Inequality valid only when terms are REAL and POSITIVE

↳ Equality holds when all terms are same.

Ques If $a+b+c=3$ & a,b,c are positive real numbers, find maximum value of $a^2b^3c^2$.

Sol'n $\frac{a}{2}, \frac{a}{2}, \frac{b}{3}, \frac{b}{3}, \frac{b}{3}, \frac{c}{2}, \frac{c}{2}$

$\therefore AM \geq GM$

$\frac{a+b+c}{7} \geq \left(\frac{a^2b^3c^2}{2^4 \cdot 3^2}\right)^{\frac{1}{7}}$

$\Rightarrow \frac{3^7}{7^7} \geq \frac{a^2b^3c^2}{2^4 \cdot 3^2}$

$\Rightarrow a^2b^3c^2 \leq \frac{2^4 \cdot 3^9}{7^7}$

$\Rightarrow \boxed{a^2b^3c^2|_{max} = \frac{2^4 \cdot 29}{7^7}}$ **Ans**

Ques Find the min. value of $f(x) = a^{a^x} + a^{1-a^x}$ where $a, x \in R$ and $a > 0$.

Soln $f(x) = a^{a^x} + a^{1-a^x}$

Let, $f(x)$ be min. at some ~~x~~ $a^x = t$

$$f(t)_{min} = a^t + a^{1-t}$$

$\because$ AM $\geq$ GM

$$\frac{a^t - a^{1-t}}{2} \geq (a^t \cdot a^{1-t})^{1/2}$$

$$a^t - a^{1-t} \geq 2\sqrt{a}$$

i.e $\boxed{\left(a^{a^x} + a^{1-a^x}\right)_{min} = 2\sqrt{a}}$ Ans

Ques Let $f: R \to R$ be such that for all $x \in R$ $(2^{x+1} + 2^{1-x})$, $f(x)$ and $(3^x + 3^{-x})$ are in AP, then find min. value of $f(x)$.

Soln $f(x) = \dfrac{(2^{x+1} + 2^{1-x}) + (3^x + 3^{-x})}{2}$

Now,

$$\frac{2^{x+1} + 2^{1-x}}{2} \geq (2^{x+1} \cdot 2^{1-x})^{1/2}$$

$$2^{x+1} + 2^{1-x} \geq 2(2)$$

$$\Rightarrow \boxed{(2^{x+1} + 2^{1-x})_{min} = 4}$$

And,

$$\frac{3^x + 3^{-x}}{2} \geq (3^x \cdot 3^{-x})^{1/2}$$

$$\Rightarrow \boxed{(3^x + 3^{-x})_{min} = 2}$$

Thus,

$$\boxed{f(x)|_{min.} = 3}$$ Ans